

Research Proposal: Evaluating Invisible Playmakers in Soccer via Structural and Probabilistic Metrics

Introduction

Traditional soccer ratings focus on goals and assists, which **overvalue strikers and undervalue deep-lying playmakers and defenders** ¹ ². This “goalscorer bias” (the **FIFA paradox**) means players who stabilize possession, manipulate space, or break defensive lines often get little credit. Recent analytics models (xT, VAEP, EPV) capture **some** hidden value by measuring expected threat or scoring probability ³ ⁴. However, they do not explicitly quantify a player’s influence on team *structure* or decision quality. We propose a novel multi-dimensional framework – the **Invisible Playmaker Index (IPI)** – that combines action-based value with structural and tactical metrics to fully credit “invisible” contributions.

Key motivations:

- Defensive midfielders and ball-retainers (Rodri, Busquets, Kroos, etc.) are **critically important** but “rarely appreciated by traditional metrics” ¹.
- Existing models capture change in scoring chances (ActionValue) ⁵, but miss effects like **organizing team shape** or **breaking opponents’ structure**.
- We hypothesize that **EntropyReduction**, **DefensiveDisruption**, and **DecisionQuality** are additional components of value. Combining them with ActionValue yields a **more comprehensive player rating**.

Related Work

Modern analytics provide tools that we will build upon:

- **Expected Threat (xT)** – A Markov-chain model assigning each pitch cell a “threat” value; passes are valued by how they move the ball into more dangerous zones ⁶ ⁷. This rewards line-breaking passes (e.g. Ozil’s through ball) even without an assist ⁷.
- **VAEP (Valuing Actions by Estimating Probabilities)** – Uses machine learning to estimate how each on-ball action (pass, dribble, tackle, shot) changes immediate goal/concede probability ⁴ ³. VAEP’s top contributors differ from goalscorers, highlighting creative midfielders ³.
- **Expected Possession Value (EPV)** – Deep models (Fernandez *et al.*) that use tracking data to estimate the expected outcome of a possession at any moment ⁸. These capture context (player positions, spacing) to rate actions.
- **Pitch Control Models** – Physics-based models (Shaw, Spearman) that compute which team can reach each spot first ⁸. These yield “control surfaces” and allow safe-pass graphs, used in some teams’ analysis.
- **Possession Plus-Minus** – Regression models (Bajons & Hornik) that use possession sequences to separate on-ball vs off-ball player ratings ⁹. These consider team context but not spatial structure.

- **GNN-Based Evaluation** – Recent work (Jiang *et al.*) models players and passes as a graph, identifying “hidden pivotal” players who link defense and attack ¹.

Despite these advances, **no existing model explicitly measures a player’s impact on team shape or decision optimality**. Our framework integrates these novel components.

Proposed Methodology

We define the **Invisible Playmaker Index (IPI)** for each player as a weighted sum of four components:

$$\text{IPI} = \alpha \times \text{ActionValue} + \beta \times \text{EntropyReduction} + \gamma \times \text{DefensiveDisruption} + \delta \times \text{DecisionQuality}.$$

Below we detail each term.

1. ActionValue (Probabilistic Impact)

This is the change in scoring opportunity due to an action. Formally, if S_i is the game state before action a and S_f is after, we set

$$\text{ActionValue} = V(S_f) - V(S_i),$$

where $V(S)$ is the probability of scoring from state S . For example, using an xT or Markov model, $V(S)$ could be the probability of eventual goal given ball position and context ⁵. The contribution of action a is thus $\Pr(\text{goal}|S_f) - \Pr(\text{goal}|S_i)$. We sum each player’s ActionValue contributions over all their actions ⁵.

2. EntropyReduction (Structural Stability)

We treat the team’s formation as a spatial probability distribution of players. Let $p(x)$ be the density of players in region x . The Shannon entropy $H = -\sum_x p(x) \ln p(x)$ measures spatial disorder. We define EntropyReduction as the decrease in team entropy caused by a player’s action:

$$\text{EntropyReduction} = H(\text{before}) - H(\text{after}).$$

A **positive** value means the team’s shape became more organized. For instance, a midfielder’s pass that draws teammates into a tight formation would lower entropy. While entropy has been used to analyze leagues’ dynamics ¹⁰, using it at the possession level to rate players is novel. This metric credits players who **stabilize or organize their team’s formation** when attacking.

3. DefensiveDisruption (Opponent Instability)

Here we quantify how a player’s action disturbs the opponent’s defensive shape. For a given defending team, we model their formation (e.g. convex hull or clustering of defensive players). Let D_{before} be a measure of defensive compactness before the action (e.g. area of convex hull), and D_{after} after the action. Then

$$\text{DefensiveDisruption} = D_{\text{after}} - D_{\text{before}}.$$

Large positive values indicate the defense was pulled apart. For example, a line-breaking through-pass that forces defenders to spread out would increase this metric. (Similar to **Pressing Game** models which compute “disruption surfaces” ¹¹, we measure the actual change in defensive structure per action.) This rewards players who **break lines and create space by disrupting the opponent**.

4. DecisionQuality (Tactical Intelligence)

We evaluate how optimal a player’s chosen action is compared to alternatives. At each decision point, consider all feasible actions (pass targets, dribble directions, shoot, etc.). For each option o , compute its estimated ActionValue V_o . Let a^* be the chosen action with value V_{a^*} , and $V_{\max}$ the maximum over all options. Then define

$$\text{DecisionQuality} = V_{a^*} - V_{\max}.$$

This will be ≤ 0 , with 0 indicating an optimal choice. (Alternatively, we may take $-(V_{\max} - V_{a^*})$ so that higher is better.) This metric credits players for consistently choosing high-value actions, capturing **game intelligence**. For example, a midfielder who picks the threat-maximizing pass (leading to maximal $\Pr(\text{goal})$) would score well. Decision-making analysis has been explored in sports, but rarely integrated into player value models, making this approach novel.

Parameter Weights

We will learn or set weights $\alpha, \beta, \gamma, \delta$ to balance the four components. For instance, $\alpha = 0.4, \beta = 0.3, \gamma = 0.2, \delta = 0.1$ might be an initial guess, but we will refine these via regression or validation experiments.

Data and Implementation Plan

- **Data sources:** We will use open soccer event data (e.g. StatsBomb Open Data) and, if possible, tracking data (Metrica or Wyscout) for pitch control. StatsBomb 360 data can serve as a proxy for tracking.
- **ActionValue model:** We will implement a Markov chain or use xT lookup tables ⁵ to compute $V(S)$. Shots/turnovers are absorbing states.
- **Entropy computation:** For each possession, use player coordinates (from freeze-frames) to estimate a density map (e.g. Gaussian kernel over players) and compute entropy before and after each action.
- **Defensive disruption:** Identify defenders’ positions before/after each attacking action. Compute a compactness metric (e.g. convex hull area) to quantify the defensive shape.
- **DecisionQuality:** At each possession state, generate possible actions (other teammate passes, dribbles). Estimate each option’s ActionValue via our model; compare the chosen action’s value to the best possible.
- **Tooling:** Python with pandas, NumPy, SciPy for calculations; `scikit-learn` or custom code for regressions; `networkx` or plotting for formations.

We will test the computation on **sample matches** (e.g. 2018 World Cup games) to refine and validate intermediate outputs (e.g. compare IPI for well-known playmakers vs raw goals/assists).

Evaluation Experiments

1. **Correlation with success:** Aggregate team IPI (sum of player IPI in games) and test correlation with outcomes (league points, goal differential). Expect stronger correlation than raw goals or xT alone.
2. **Comparison to existing metrics:** Compare top-10 players by IPI to those by xT and VAEP. We expect IPI to surface deep playmakers and structural stars that xT/VAEP rank lower ³ ¹.
3. **Case studies:** Qualitatively analyze specific players. For example, calculate IPI for Rodri, Busquets, Kroos, comparing their IPI to their goals/assists tallies. We expect “invisible” players to have much higher IPI than traditional stats suggest ¹ ¹².
4. **Ablation analysis:** Remove one component at a time (e.g. only ActionValue vs full IPI) to quantify the contribution of each dimension to predictive power.

Expected Contributions

This research will deliver: - **A novel combined metric (IPI)** capturing both probabilistic and structural contributions.

- **Mathematical definitions** for EntropyReduction and DefensiveDisruption, which have not been used in prior player ratings.

- Empirical evidence that IPI explains team success and identifies undervalued players better than existing metrics ³ ¹².

Such a metric could transform scouting and analysis by recognizing crucial playmakers who work off the stat sheet.

Timeline

1. **Months 1–2:** Data collection (download event and tracking data) and literature review of xT/VAEP/EPV models ⁴ ⁸.
2. **Months 3–4:** Develop ActionValue and DecisionQuality modules; compute initial IPI components on a few matches.
3. **Months 5–6:** Implement EntropyReduction and DefensiveDisruption metrics. Validate each component’s reasonableness.
4. **Months 7–8:** Run full experiments: compute IPI for all players, perform correlation and ranking analyses.
5. **Months 9–10:** Writing and refinement of the paper. Prepare figures (e.g. IPI vs goals plots) and submit to a sports analytics venue.

References

Key sources that inspired this work include: - Shaw & Spearman’s pitch control foundation (footnotes in ¹³).

- Decroos *et al.* (2019) on VAEP ⁴ ³.

- Singh’s xT framework ⁷ ⁶.

- Fernandez *et al.* on EPV ⁸.

- Bojinov & Bornn (2016) on defensive disruption surfaces ¹¹.

- Elsharkawi *et al.* (2025) on integrating xG, xT, and defense ¹².

These form the foundation; our proposal extends them by adding the entropy, disruption, and decision components.

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